Complete Workflow Architecture

End-to-end system view. Follow the numbered flow, then use the lower panels for component and storage detail.

1 **Raw energy data**
UCI household CSV
Date, time, 7 meters

2 **Data ingestion**
Read semicolon CSV
Combine date + time

3 **Clean + validate**
Timestamps and numbers
Remove duplicates, fill gaps

4 **Time-series prep**
Datetime index, minute rows
Hourly means and kWh fields

5 **Analytics + AI engine**
Energy analytics
Anomaly detection
Forecasting + evaluation

6 **Structured results**
Metrics, trends, evidence
Forecasts, caveats, trace

7 **API / backend**
FastAPI routes
Validate and return JSON

8 **AI assistant**
Understand question
Select evidence + explain

9 **Web dashboard**
Charts, tables, AI chat
Date-scoped pages

10 **User insights**
Review evidence
Plan next steps

DATA INPUT, QUALITY AND PREPARATION

Source columns

Date + Time; Global Active Power
Global Reactive Power; Voltage
Global Intensity; Sub-metering 1-3
Raw source file is optional at runtime

Cleaning rules

Drop invalid timestamps; keep last duplicate
Coerce seven meter columns to numeric
Interpolate internal gaps up to 5

Quality report + API checks

Loaded/cleaned records, missing values, duplicates, invalid times
Date coverage and applied cleaning rules; API reports 1-minute sampling
Runtime validates schema, numeric data, timestamps, and date bounds

Time-series preparation

Datetime index; hour/weekday/date/month groupings; hourly means
Active energy = sum / 60; sub-meter Wh = hourly sum / 1000; Other = residual

ANALYTICS AND MACHINE-LEARNING MODULES

Energy analytics

Mean, max, P95, peak timestamp and hours
Weekday profile; daily and monthly trends
Total kWh; estimated sub-meter mix
Voltage, current, and reactive-power means

Anomaly detection

Weekday-hour median + robust MAD score
Isolation Forest candidate signal
Combine candidates; rank by MAD score
Unusual patterns are review candidates

Forecasting

Check continuity and history sufficiency
168-hour weekly, 24-hour daily fallback
Recent median / insufficient-history fallback
Bound horizon to 1-168 hours; stale warning

Forecast evaluation

Chronological train/test holdout
MAE, RMSE, MAPE; minimum 48 hours
MAPE denominator floor: 0.1 kW
Evaluation baseline is computed separately

APPLICATION AND USER EXPERIENCE

API / backend

GET summary, trends, anomalies, forecast, evaluation, quality, search
POST /api/ask; date/parameter validation; HTTP error handling
JSON responses connect React pages to Python calculations

AI Energy Intelligence Assistant

Rule-based query understanding; identify intent and parameters
Call requested backend calculations; rank keyword evidence
Build factual answer + trace; optional LLM phrasing is grounding-checked

Web dashboard

Overview / Analytics / Anomalies / Forecast / Insights / Reports
Data Quality / Evaluation / AI Chat; renders charts and tables
Responses and feedback are shown in the browser

DATA STORAGE

Raw

Original semicolon CSV
Not bundled; optional for preprocessing

Processed

Cleaned minute CSV
Hourly DataFrame is derived in memory

Metadata

Quality JSON
Cleaning rules and source date coverage

Results

Returned as API JSON
No results database
Chat evaluation in browser

GUARDRAILS AND RELIABILITY

- Missing / invalid data checks
- Duplicate and date validation
- History and continuity checks
- Stale forecast warning
- Anomaly API cap: 200
- Dashboard requests up to 100
- MAPE floor: 0.1 kW
- Chronological holdout + errors

Limits shown match the implementation; anomaly flags are not confirmed faults.

Runtime note: the included cleaned CSV is read by FastAPI; agent modules run in-process and are not separate services.

Analytics and AI Engine - Technical Detail

The four modules share date-scoped hourly data. Branch steps and limits below follow the Python implementation.

A. ENERGY ANALYTICS

Input

Selected hourly data
Global_active_power
Sub-meter energy fields

Demand statistics

Mean active power; maximum hourly mean and timestamp
P95 demand; minimum; top three clock-hour averages
Weekday profile; daily and monthly trends; daily change
Total estimated kWh; sub-meter 1/2/3 + Other residual
Voltage, current (Global Intensity), reactive-power means

Energy insights

Structured summary, hourly profiles, daily/monthly series, evidence strings
Used by dashboard pages, Copilot, composed insight and decision routes

API

JSON metrics

B. ANOMALY DETECTION

Hourly demand

Apply selected dates
Drop missing power

Weekday + hour

Group matching slots
Calculate slot median

Robust deviation

MAD x 1.4826
Flag |score| >= 4.5

Isolation Forest

Cached hourly model
Separate candidate flag

Combine and rank

Candidate = MAD threshold OR Isolation Forest flag
Sort candidates by absolute MAD score
Return observed, baseline, score, and prior-hour/day/week values

Result limit and meaning

API accepts limit 1-200; direct function caps at 200
Dashboard currently requests up to 100 results
Anomaly result is a statistical review candidate, not a fault

Fallback: if Isolation Forest cannot run, use weekday-hour MAD alone.

C. FORECASTING

Select history

Filter date range
Exclude hours <45 minutes
Require at least 3 points

History continuity

168 continuous hours?
Else 24 continuous hours?
Else use median fallback

Fallback hierarchy

1. Weekly seasonal-naive: repeat prior 168-hour pattern
2. Daily seasonal: repeat prior 24-hour pattern
3. Recent median when >=24 hourly points
4. Insufficient-history placeholder below that

Forecast output and safeguards

Generate next N hours; horizon is bounded to 1-168 hours
Return model, fallback_used, history_end, mean and hourly values
Warn if latest selected reading is older than 7 days
Caveat: historical baseline only; no weather or occupancy inputs

D. FORECAST EVALUATION

Historical series

Selected hourly power
At least 48 hours required

Chronological split

Final 24 to 672 hours held out
Train only on earlier values
No future-data leakage

Seasonal-naive predictions

168-hour lag if training >=168 hours
Otherwise 24-hour lag

Evaluation metrics

MAE: average absolute error | RMSE: error with larger misses weighted more
MAPE: percentage error; denominator is floored at 0.1 kW near zero
Forecast evaluation is a separate holdout calculation, not the live forecast output

API, Assistant and Dashboard Workflow

A request enters FastAPI, the Copilot selects supported local evidence, and React displays structured results.

API / APPLICATION LAYER

Request path

GET focused analytics endpoints
POST /api/ask(question, dates, context)
Validate request length and date parameters
Apply topic, date and context guardrails

Response shape

Endpoint-specific JSON metrics
Copilot: answer + evidence + trace
NLP interpretation + search results
Forecast model and caveats
No results database; results returned to caller

Important routes

GET /api/summary, /trends, /anomalies, /forecast, /evaluation
GET /api/data-quality, /search, /agents/insight, /agents/decision
POST /api/ask; FastAPI also serves the built React app

AI ENERGY INTELLIGENCE ASSISTANT

Question understanding

Rule-based token normalization, synonyms, intents and parameter extraction
Supported focus: date scope, sub-meter, measurement, weekday, forecast horizon
Guardrails ask for clarification when the dataset cannot answer the request
Example: "What were the peak demand hours?"
Example: "Did anything unusual happen yesterday?"
Example: "What is the expected demand for the next 24 hours?"
Example: "How much energy was consumed this month?"

Evidence-first answer

Call only the requested calculations; rank keyword evidence; compose facts + limits + trace

Optional language model

No key = local answer; configured model may rephrase; unsupported numbers are rejected

FRONTEND / PRESENTATION LAYER

Dashboard sections

Overview: average, max, total energy, P95
Analytics: demand, daily/monthly, peak hours
Anomalies: time, observed, baseline, score
Forecast: history, values, model,

More dashboard sections

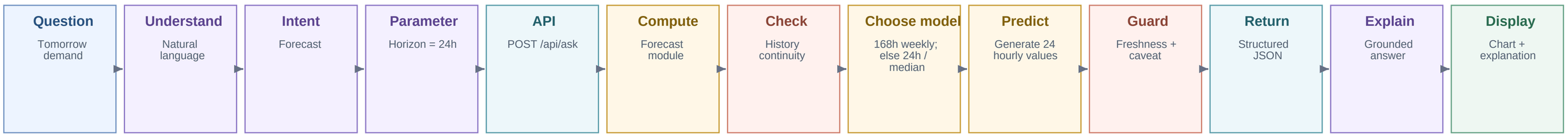
Data Quality: records, missing, duplicates, rate
AI Chat: question, answer, evidence, rating
Insights, reports and CSV export
Date scope passed to backend endpoints

How results are shown

React pages call the JSON API through services/api.ts
Charts, tables, metrics and evidence panels render returned values
Chat evaluation and feedback are saved in browser localStorage

EXAMPLE END-TO-END QUERY FLOW

User asks: "What is the expected electricity demand for tomorrow?"



GUARDRAILS AND RELIABILITY

Schema / missing values	Invalid timestamps / duplicates	Date coverage + timezone	Min 3 forecast points	168h / 24h continuity	Horizon <=168 hours
Stale data >7 days	Anomaly limit <=200	Chronological holdout	MAPE floor 0.1 kW	Grounded numeric response	HTTP error handling

The source is one household of historical data. No live grid, weather, occupancy, appliance events or verified fault labels are present.